

Aritra Ray

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PROFESSIONAL SUMMARY

Boston College graduate with dual **MS** in **Applied Analytics** and **Applied Economics** and **UG** in **Computer Science Engineering**. Experienced in end-to-end AI and data engineering, LLM evaluation, prompt engineering, and **Python**-based pipeline development. Combines **SQL**, **statistical rigor**, with stakeholder communication to turn complex data into actionable business decisions.

TECHNICAL SKILLS

- **Languages and Concepts:** Python, R, SQL, Microsoft Excel, statistical modeling, Power Query Editor
- **Data Engineering and Quality:** ETL/ELT (pandas, NumPy), Streamlit, API integration, data validation, schema design
- **BI and Visualization:** Tableau, Power BI, matplotlib, dashboard generation, KPI reporting
- **Modeling and Cloud:** Regression, Forecasting, A/B testing, Hypothesis testing, Machine Learning, scikit-learn, AWS, GCP

WORK EXPERIENCE

Fiscal Data Analyst **Chestnut Hill, MA**
Boston College, Woods College of Advancing Studies *Aug 2023 – Dec 2025*

- Built an automated **Streamlit** pipeline from daily enrollment snapshots across 20+ programs, surfacing **KPIs** for leadership.
- Used **Tableau** to analyze structured enrollment data, identifying trends in yield, intake, and program preferences.
- Modeled revenue and enrollment projections tracking **YoY growth** and **cross-semester deltas** to quantify the **yield rate**.

AI Engineer Intern **Kolkata, India**
Finurja *Feb 2024 – Apr 2024*

- Wrote a **pandas** script to normalize bank statements across 3+ formats into one schema, enabling consistent LLM classification.
- Compared **GPT-4 (82.8%)** and Gemini Pro (79.5%) against a rule-based keyword classifier (52%) on transaction classification.
- Identified **80-token prompts** as the accuracy-maximizing configuration via controlled prompt-length **A/B testing**.

Business Intelligence Data Analyst **Kolkata, India**
Ernst & Young Global Delivery Services *Jun 2023 – Aug 2023*

- Automated 12 infrastructure security audit tests using **Python** and **Power Query**, cutting execution effort by **75%+**.
- Consolidated security audit results into **Power BI dashboards**, replacing fragmented Excel reporting across teams.
- Implemented **rule-based exception** logic to automatically **flag non-compliant access** configurations across all tests.

PROJECT & INTERNSHIP EXPERIENCE

End-to-End Search Relevance Optimization - Information Retrieval **Chestnut Hill, MA**
SurveyMonkey *November 2025*

- Scraped **500+ help pages**, created and evaluated 78 labeled queries, exposing **45% search failure rate**.
- Lifted search success from **~55% to 77.5%** on unseen queries by shipping hybrid retrieval ranking method with RRF fusion.
- Replaced CrossEncoder (40-60s per batch) with precomputed MiniLM embeddings, cutting **re-rank latency** to near-instant.

Statistical Modeling - Hurricane Sandy Property Value Impact Analysis **Chestnut Hill, MA**
Boston College, Woods College of Advancing Studies *March 2025*

- Applied **Difference-in-Differences** to **8,000+ NJ property transactions** using **FEMA** damage intensity
- Merged MOD-IV, FEMA damage, and Census data to construct a clean **panel dataset** for causal estimation.
- Estimated an increase of **\$53,760; placebo test** on a false cutoff and event-study confirmed the effect was Sandy-specific.

FMCG Retail Shopper Loyalty & Revenue Forecasting **Chestnut Hill, MA**
Boston College, Woods College of Advancing Studies *December 2024*

- Analyzed 3+ years of pet food transactions (2019 - 2022) in **R** to quantify market share shifts and shopper behavior.
- Defined 3 loyalty tiers on frequency, recency, and LTV; quantified brand switching rates and modeled patterns.
- Forecasted monthly revenue using **ETS**, **ARIMA**, and **Seasonal Naive**; selected best model via cross-validated **RMSE**.

EDUCATION

Boston College **Chestnut Hill, MA**
Dual MS in Applied Analytics & Applied Economics | **GPA 3.80 / 4.00** *December 2025*
Institute of Engineering & Management **Kolkata, India**
Bachelor of Technology in Computer Science Engineering | **GPA 3.93 / 4.00** *May 2023*

PUBLICATION & RESEARCH

- Built and evaluated a stateless **LSTM**-based pipeline for **speech and text emotion detection**, achieving **~86%** accuracy.
- Presented findings to stakeholders and published the work in **Frontiers of ICT in Healthcare (2023)**.